EMMET: A Potential-Field Routing Heuristic with Adaptive Congestion Thermostat and Loss Memory

Carlos López
Independent Researcher
carlos@codix.cat
github.com/Carloscodix/EMMET

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Abstract

We present EMMET, an adaptive routing algorithm grounded in classical mechanics and thermodynamics. Network packets are modeled as physical particles navigating a composite potential field with three terms: distance to destination, instantaneous link congestion, and decayed historical packet loss. The TTL is interpreted as physical energy; packets that exhaust their budget are considered to undergo thermal dissipation. The loss term carries a frozen warm-up snapshot subject to half-life decay, modeling persistent thermal memory of past failures. An ε-greedy stochastic term implements local exploration. A global congestion thermostat adapts the congestion weight β according to the mean normalized network load.

We characterize a phase transition in the algorithm's behavior as a function of network density: below a critical threshold $\rho_c \in [0.15, 0.30]$ on Erdős–Rényi graphs, the potential field collapses into topological dead ends. Above the threshold, EMMET achieves near-zero packet loss in the dense regime (mean below 0.2 packets per 200-step run for $\rho \geq 0.30$ on $n = 20$). We further identify a β sweet spot at $\beta = 3.5$–4.0 followed by a field saturation regime, and the empirical relationship between mechanisms: thermal memory and adaptive β are **synergistic**, while local lookahead h=2 and thermal memory are **substitutes**.

EMMET is validated against three baselines (Shortest Path, ECMP, LASP) on synthetic Erdős–Rényi graphs ($N \in \{20, 50, 100\}$) and real Internet topologies (Abilene, GEANT) from the Internet Topology Zoo, across three batteries totaling 28,800 simulations (100 seeds per scenario for $n = 20$, $n = 50$, and real topologies; 50 seeds for $n = 100$ due to compute cost). The full system (adaptive β + thermal) reduces packet loss by 55 % to 65 % versus LASP on synthetic networks and 54.5 % on GEANT, with a 12.3 % reduction on the small Abilene topology that previous variants could not improve. Three independent adversarial code audits identified and corrected methodological errors.

Keywords: adaptive routing, potential field, phase transition, thermal dynamics, complex networks, network science.

1 Introduction

The standard approach to network routing — minimize path length under some cost metric — treats the network as a static optimization problem. The packet follows the shortest path and arrives, or it does not.

Real networks are dynamic systems. Links congest, degrade, and recover. A packet that blindly follows the shortest path may arrive at a saturated link and be dropped, while an alternative longer path delivers reliably.

This paper asks: *what if a packet behaved like a physical particle?* In classical mechanics, a particle moving through a field does not follow the straight-line path — it follows the path of least potential energy, trading distance for the avoidance of high-energy regions. Friction dissipates motion. Heat marks regions of maximum energy exchange. A particle with finite energy that cannot reach its destination dies of thermal dissipation.

This framing produces a concrete algorithm with measurable physical analogues. We name it EMMET (Energy-Minimizing Multi-dimensional Edge-based Traversal). Each routing decision minimizes a composite potential combining distance, congestion, and loss memory, under a globally adapted congestion weight, with stochastic exploration to break greedy rigidity.

Our contributions are:

1. A potential-field routing algorithm with four physically motivated mechanisms: warm-up loss snapshot, half-life decay, ε-greedy exploration, and adaptive β thermostat.
2. Empirical characterization of a critical density threshold below which the potential field collapses (phase transition).
3. Identification of a β sweet spot followed by a field saturation regime.
4. Empirical demonstration that thermal memory and adaptive β are synergistic, while lookahead h=2 substitutes for thermal memory.
5. Validation across 28,800 simulations on synthetic and real Internet topologies, after three adversarial code audits.

2 Related Work

Potential-field routing has been studied for wireless sensor and ad-hoc networks. Lenders et al. [1] propose density-based anycast with explicit analysis of local maxima, deriving theoretical bounds for local-minima-free operation under a single-term potential. Their analysis does not characterize a critical density for composite multi-term potentials.

Geographic routing literature [3] documents local minima in greedy forwarding and addresses them through recovery mechanisms (face routing, perimeter routing) rather than characterizing the failure regime.

Backpressure routing [2] routes on queue differentials and is throughput-optimal but degrades latency.

ECMP [4] distributes flows hash-based among equal-cost paths and is the industry standard for load balancing in IP networks. **LASP** (Load-Aware Shortest Path) extends Dijkstra with congestion-aware weights and is a standard traffic-engineering baseline.

None of these works characterize a critical density threshold for a composite three-term potential function combining distance, congestion, and decayed loss memory; nor do they study mechanism interactions (synergy vs. substitution) between exploration, lookahead, and adaptive parameters. This is the gap EMMET addresses.

3 The EMMET Model

3.1 Network Model

The network is an undirected graph $G = (V, E)$ where each edge (u, v) carries:

- latency(u, v) — static propagation delay
- capacity(u, v) — maximum simultaneous load before a packet is dropped
- load(u, v) — current traffic level
- loss(u, v) — accumulated packet loss count

Loss and load dynamics. At each simulation step a single packet is injected with a random source–destination pair. The packet traverses the chosen path edge by edge. Before each edge transit $\mathrm{load}(u, v) \leftarrow \mathrm{load}(u, v) + 1$; if the new load exceeds $\mathrm{capacity}(u, v)$, the packet is dropped at that edge, $\mathrm{loss}(u, v)$ is incremented by one, and no further hops are attempted. There are no queues or retransmissions. After each step, load on every edge decays by a factor of 0.9, modelling traffic that is bursty rather than permanently saturating. This idealisation isolates the effect of the routing decision on loss, at the cost of not modelling sustained flows; we discuss this limitation in §6.4.

3.2 Potential Function

For a packet at node u considering neighbor v toward destination dst:

$$P(u, v, \mathrm{dst}) = \alpha \cdot d(v, \mathrm{dst}) + \beta_{\mathrm{eff}} \cdot \frac{\mathrm{load}(u, v)}{\mathrm{capacity}(u, v)} + \gamma \cdot \mathrm{loss\_snapshot}(u, v) \tag{1}$$

where α and γ are fixed weights, and β_{eff} is governed by the adaptive thermostat (§3.6).

3.3 Routing Rule

At each hop, the packet moves to the neighbor minimizing P, excluding previously visited nodes. With probability ε, the second-best neighbor is chosen instead — this implements stochastic exploration (§3.7).

3.4 TTL as Physical Energy

A packet is born with energy budget $E = \mathrm{TTL\_FACTOR} \cdot |V|$ hops. Two distinct termination conditions exist:

- **dead_end**: no unvisited neighbors — topological local minimum
- **ttl_expired**: energy exhausted — thermal dissipation

These are tracked separately as failure modes signaling field collapse versus insufficient energy for the topology.

3.5 Thermal Memory: Snapshot, Decay, and Warm-up

The loss_snapshot term carries persistent information about past failures. It is constructed in three steps:

Warm-up phase. Before measurement, EMMET routes $\mathrm{warmup\_steps} = \max(20, |V| \cdot 5)$ packets and freezes the resulting per-edge loss values into a snapshot dictionary.

Read-only during measurement. The snapshot is consulted at every routing decision but is not updated by losses occurring during measurement. This eliminates information asymmetry against the baselines.

Half-life decay. After each step, $\mathrm{snapshot}(u, v) \leftarrow \mathrm{decay} \cdot \mathrm{snapshot}(u, v)$, with $\mathrm{decay} = \exp(-\ln 2/\mathrm{HALF\_LIFE})$. For $\mathrm{HALF\_LIFE} = 100$ steps, $\mathrm{decay} \approx 0.9931$ — the snapshot loses 50 % of its value every 100 steps. This models heat dissipation.

3.6 Adaptive β Thermostat

A fixed β assumes uniform congestion across the network. Real topologies have heterogeneously stressed regions. The adaptive thermostat scales β with the global thermal state:

$$\beta_{\mathrm{eff}} = \beta_{\mathrm{base}} \cdot \left(1 + \theta \cdot \left\langle \frac{\mathrm{load}(u, v)}{\mathrm{capacity}(u, v)} \right\rangle \right) \tag{2}$$

where $\langle\cdot\rangle$ denotes the mean over all edges and $\theta \geq 0$ is the sensitivity. With $\theta = 0$, β is fixed; with $\theta > 0$, β scales up under network stress. The thermostat introduces no learning state — it is a stateless feedback mechanism evaluated at routing time.

3.7 ε-greedy Stochastic Exploration

With probability ε, the routing decision selects the second-best neighbor by potential ranking instead of the best. This breaks greedy rigidity, enables exploration of locally suboptimal paths, and provides robustness to small modeling errors. We use $\varepsilon = 0.10$.

In an early implementation, an additive constant offset was used as an "exploration term"; this is mathematically inert (it adds equally to all neighbors and does not change the argmin), and was identified by an adversarial audit. The ε-greedy formulation replaces it with a real exploration mechanism.

3.8 Implicit Momentum

The visited set implements infinite inertia by construction — a previously visited node is never reconsidered, equivalent to assigning it infinite potential. Empirical tests with explicit momentum terms confirmed no measurable additional effect, so explicit momentum is omitted.

4 Experimental Setup

4.1 Topologies

- **Synthetic**: Erdős–Rényi $G(N, p)$ with size-dependent density coverage:
  - $N = 20$: $p \in \{0.05, 0.10, 0.15, 0.20, 0.25, 0.30, 0.40, 0.50\}$
  - $N = 50$: $p \in \{0.05, 0.10, 0.15, 0.20, 0.25, 0.30\}$
  - $N = 100$: $p \in \{0.05, 0.10, 0.15, 0.20\}$ (sparse regime only, where the phase transition is observable; denser n=100 graphs converge trivially to zero loss for all strategies)
- **Real**: Abilene ($N = 11$, $\rho = 0.25$) and GEANT ($N = 40$, $\rho = 0.08$) from the Internet Topology Zoo [5]

4.2 Traffic Model

For each scenario and seed, 200 random source–destination pairs are generated from a fixed traffic seed, independent of any routing strategy's randomness. All strategies see identical traffic.

4.3 Statistical Validation

100 independent seeds per scenario for synthetic n=20, n=50, and real topologies; 50 seeds for n=100 due to computational cost. Mean and standard deviation reported throughout. Across the three batteries (canonical 4-strategy, lookahead h=2 with 6 strategies, and adaptive-β with 6 strategies), 28,800 simulations were executed ($1800 \times 4 + 1800 \times 6 + 1800 \times 6$).

4.4 Baselines

- **SP**: Dijkstra on latency
- **ECMP**: random choice among shortest paths [4]
- **LASP**: Dijkstra on latency $\cdot$ (1 + load/capacity)

4.5 Audit-Clean Implementation

Three independent adversarial code reviews identified and corrected:

- RNG contamination across strategies (separate RNGs for traffic, baseline path choice, and EMMET exploration)
- Asymmetric information in EMMET's loss term (snapshot is read-only during measurement; populated only by warm-up)
- Decay scale mismatch (switched from 0.95^{step} to half-life formulation)
- Mathematically inert exploration constant (replaced with ε-greedy)
- Non-portable absolute file paths (replaced with portable `Path(__file__).parents[1]`)
- Latency metric mixing partial work of lost packets (separated into latency-per-delivered and latency-per-attempt)

All reported results are from the audited implementation.

4.6 Selection-Bias Verification

The metric *latency per delivered packet* is well-defined within each strategy but, when compared across strategies that deliver different subsets of packets, is exposed to selection bias: a strategy that drops more packets may show artificially low mean latency because the surviving packets are not a random sample. To rule out this artefact, we re-ran a focused experiment on $\text{ER}(20, p)$ for $p \in \{0.05, 0.10, 0.15, 0.20, 0.25, 0.30\}$ with 50 seeds per density, recording per-packet outcomes for SP, LASP and EMMET full. We then compared two metrics: (i) the original mean latency over each strategy's own delivered set, and (ii) the *conditional* mean latency over the subset of packets delivered by all three strategies (the common delivery set). The two metrics differ by less than 0.9 latency units across all six densities, and the strategy ordering (SP fastest, LASP intermediate, EMMET slowest) is identical under both metrics. Full per-density numbers are in `data/selection_bias_analysis.json`. We therefore retain *latency per delivered packet* as the headline latency metric.

4.7 Default Parameters

$\alpha = 1.0$, $\beta_{\text{base}} = 3.0$, $\gamma = 2.0$, $\varepsilon = 0.10$, $\theta = 1.0$, TTL_FACTOR = 2, HALF_LIFE = 100 steps.

5 Results

5.1 Density Sweep: Phase Transition

Density swept on Erdős–Rényi $G(20, p)$ from p=0.05 to p=0.50 (Table 1).

Finding 1 — Phase transition at $\rho_c \in [0.15, 0.30]$. Below this range, graphs are rarely connected; above $\rho = 0.30$, graphs are 100 % connected and EMMET achieves a mean loss below 0.2 packets per 200-step run.

Finding 2 — Loss reduction at low density, with a latency cost. At $p < 0.15$, EMMET reduces losses by 54–67 % versus SP. SP retains a latency advantage of 3–13 % (lower mean latency per delivered packet, verified under both per-strategy and conditional-on-common-delivery metrics; see §4.6 and `data/selection_bias_analysis.json`). EMMET and SP thus occupy distinct corners of the loss-versus-latency trade-off space; they are not strictly comparable.

Finding 3 — Dead ends as field collapse signature. Dead-end counts drop sharply at $p \approx 0.25$–0.30, coinciding with full topological connectivity.

Table 1: Density sweep on $G(20, p)$, 100 seeds per point.

Density	SP loss	LASP loss	EMMET full loss	Connectivity
0.05	11.98	11.53	**5.15**	0 %
0.10	39.67	35.14	**12.30**	7 %
0.15	18.81	11.48	**7.61**	43 %
0.20	10.57	4.78	**2.39**	80 %
0.25	4.57	1.14	0.41	93 %
0.30	2.45	0.53	0.17	100 %
0.40	1.07	0.18	0.04	100 %
0.50	0.83	0.11	0.00	100 %

5.2 Beta Sensitivity (Fixed β)

Beta swept on $G(20, 0.30)$ from β=0.1 to β=5.0 with γ=2.0 (Table 2).

Table 2: Beta sensitivity on $G(20, 0.30)$.

β	EMMET losses	EMMET latency
1.0	0.17	4.856
2.0	0.07	4.870
3.0	0.07	4.891
3.5	**0.00**	4.898
4.0	**0.00**	4.915
4.5	0.03	4.936
5.0	0.03	4.960

Finding 4 — β sweet spot at 3.5–4.0. Zero packet loss at minimum latency cost.

Finding 5 — Field saturation above β = 4.0. Excessive congestion aversion forces packets onto longer routes that congest previously uncongested links — the β-loss relationship is non-monotonic above the sweet spot. This motivates the adaptive thermostat (§3.6) which sets β_{eff} dynamically based on global load.

5.3 Multi-Mechanism Comparison

Six strategies evaluated on representative scenarios (Table 3).

Table 3: Multi-mechanism comparison: mean packet loss, 100 seeds (50 for n=100). Lower is better. "cold" denotes greedy potential field without snapshot; "thermal" adds warm-up + decay + ε-greedy; "adaptive" uses adaptive-β without snapshot; "full" combines adaptive-β with thermal.

Scenario	SP	LASP	cold	thermal	adaptive	full
ER ρ=0.05 (n=20)	11.98	11.53	11.25	5.46	10.96	**5.15**
ER ρ=0.10 (n=20)	39.67	35.14	33.67	13.15	32.24	**12.30**
ER ρ=0.05 (n=50)	16.87	11.70	12.82	5.27	12.10	**4.50**
Abilene	56.67	46.37	44.36	43.68	42.62	**40.65**
GEANT	26.38	11.68	11.11	6.05	9.97	**5.31**

Finding 6 — Thermal memory is the dominant single mechanism. Adding the warm-up snapshot with decay reduces losses by 46 % to 59 % over greedy-cold across the board.

Finding 7 — Adaptive β contributes additional gains synergistically with thermal memory. EMMET full (both mechanisms) outperforms either alone on every scenario tested. The full system reduces losses by 55 % to 65 % versus LASP on synthetic networks and 54.5 % on GEANT.

Finding 8 — Adaptive β is the only mechanism that improves Abilene. The small Abilene topology (N=11) was the difficult case for previous variants. Adaptive β alone reduces losses by 8.1 % versus LASP, and the full system reaches 12.3 % — the first significant improvement on this scenario.

Finding 9 — ECMP degenerates to SP without equal-cost ties. With real-valued latency, ECMP rarely finds multiple equal-cost paths and produces results identical to SP. ECMP is effective only in topologies engineered with uniform link costs (e.g., datacenter fat-trees).

5.4 Mechanism Interaction: Synergy vs. Substitution

A separate battery (`experiments/emmet_lookahead.py`, results in `data/lookahead_summary.json`) tested local lookahead with horizon h=2: at each routing decision, the potential is evaluated over the next two hops rather than one (Table 4).

Table 4: Lookahead h=2 effect on EMMET cold and thermal variants.

Scenario	cold	cold + LA2	thermal	thermal + LA2
ER ρ=0.05 (n=20)	11.25	11.37	5.46	7.85
ER ρ=0.10 (n=20)	33.67	34.10	13.15	19.13
ER ρ=0.05 (n=50)	12.82	12.18	5.27	7.68
GEANT	11.11	10.42	6.05	6.19

Finding 10 — Local lookahead does not improve EMMET; it degrades the thermal variant. Lookahead h=2 yields negligible change to EMMET cold (within ±1.0 % in n=20 sparse and modest improvement of 5–6 % in n=50 and GEANT) but consistently increases losses for EMMET thermal by 2 % to 45 %. The combination is therefore counterproductive whenever a thermal snapshot is in use. We interpret this as a substitution effect: thermal memory and local lookahead both provide non-myopic information about near-term link state, and their signals interfere when combined. The snapshot is a more efficient signal for the same purpose.

This contrasts with adaptive β, which is **complementary** to thermal memory. The distinction is meaningful: synergistic mechanisms operate on different signals (β operates on global state, thermal on per-edge history), while substitutive mechanisms compete for the same signal (both lookahead and thermal aim to anticipate failure).

5.5 Pareto Frontier Analysis

Pareto analysis (`experiments/emmet_pareto.py`, results in `data/pareto_summary.json`). Across the 20 scenarios, Pareto-optimal frequency in the (latency-per-delivered, mean-loss) plane (Table 5).

Finding 11 — SP and EMMET full occupy distinct corners of the Pareto frontier. SP is always Pareto-optimal because it minimizes latency; EMMET full is Pareto-optimal in 40 % of scenarios because it minimizes loss. The Pareto analysis confirms EMMET as the loss-minimizing extreme of a routing trade-off, not as a universal dominator.

Table 5: Pareto-optimal frequency across 20 scenarios.

Strategy	Pareto-optimal	% of scenarios
SP	20 / 20	100 %
LASP	17 / 20	85 %
EMMET cold	14 / 20	70 %
EMMET thermal	3 / 20	15 %
EMMET adaptive	8 / 20	40 %
EMMET full	8 / 20	40 %

6 Discussion

6.1 Phase Transition Interpretation

The transition at $\rho_c \in [0.15, 0.30]$ has a natural physical interpretation. Below the critical density, the graph is frequently disconnected; the potential field has no gradient to follow because the destination is unreachable from many source nodes. This is analogous to a system below its percolation threshold. Above the critical density, the field is well-defined everywhere.

6.2 Field Saturation and the Adaptive Thermostat

The non-monotonic β behavior above $\beta = 4.0$ reveals a saturation effect: overly aggressive congestion aversion forces packets onto longer paths, saturating links that would otherwise have been free. The adaptive thermostat resolves this by setting β_{eff} dynamically: high during stress, low during relaxation. This converts a parameter-tuning problem into a self-regulating mechanism.

6.3 Synergy and Substitution Among Mechanisms

The empirical distinction between synergistic and substitutive mechanisms is one of the main contributions of this work. A mechanism designer adding components to an algorithm typically assumes additive benefit; our results show this assumption fails for some mechanism pairs. The distinction appears to follow from whether the mechanisms operate on overlapping or distinct signals:

- **Adaptive β + thermal memory** — distinct signals (global load vs. per-edge history) $\rightarrow$ synergy.
- **Lookahead h=2 + thermal memory** — overlapping purpose (anticipate failure) $\rightarrow$ substitution.

This suggests a design principle: when adding a mechanism to a potential-field algorithm, identify the signal it consumes and check whether existing mechanisms already exploit that signal.

6.4 Limitations

- $N \leq 100$ nodes synthetic; real-Internet-scale topologies (> 1000 nodes) not yet measured
- Warm-up phase is sensitive to the ratio of warmup_steps to $|V|$ in very small topologies
- Static α, γ, ε; adaptive scheduling deferred to future work
- All results from simulation; no real deployment tested
- LASP is a strong but not exhaustive baseline; comparison against TE controllers (e.g. MATE, B4-style flow assignments) deferred to future work

7 Conclusion

We presented EMMET, a physics-inspired adaptive routing algorithm modeling packets as particles in a composite potential field with thermal dynamics and an adaptive global thermostat. We characterized empirically:

1. A phase transition at $\rho_c \in [0.15, 0.30]$ below which the field collapses
2. A clear loss-versus-latency trade-off across the density range, with EMMET dominating loss and SP dominating latency
3. A β sweet spot at 3.5–4.0 followed by a field saturation regime
4. Synergy between adaptive β and thermal memory
5. Substitution between local lookahead and thermal memory
6. Loss reductions of 55–65 % versus LASP on synthetic networks and 54.5 % on GEANT, with 12.3 % reduction on the small Abilene topology that previous variants could not improve.

EMMET trades a modest latency cost for substantial loss reduction. The physical framework provides interpretable parameters with clear analogues — friction, heat, energy, thermostat, exploration — and identifies failure regimes (field collapse, saturation) and mechanism interactions (synergy, substitution) that prior adaptive routing literature has not formally characterized.

Three independent adversarial code reviews validated the implementation. All data, code, and reproducibility scripts are available at the project repository under MIT license.

Code and Data Availability

All experiments are reproducible from the public repository: `https://github.com/Carloscodix/EMMET`. The repository includes implementation source, experiment scripts, raw and aggregated JSON results for all 28,800 simulations, paper-grade figures, and `requirements.txt` for environment setup. License: MIT.

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